

Rakshit Verma

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Education

Vellore Institute of Technology, Bhopal

CGPA : 8.25

Bachelor of Technology in Computer Science Engineering

Sept. 2023 – May 2027

CBSE – Class XII

85%

Maharaja Agresen Public School

2022

CBSE – Class X

84.2%

Maharaja Agresen Public School

2020

Projects

YatraMind AI | *Python, FastAPI, Streamlit, LangGraph, LangChain, Groq API*

Dec 2025 – Mar 2026

- Architected a production-ready multi-agent AI travel planner using LangGraph, FastAPI, and Groq LLM, enabling intelligent itinerary generation through six specialized autonomous agents.
- Orchestrated 5+ live APIs for flights, hotels, maps, weather, currency conversion, visa guidance, safety intelligence, and expense management to deliver end-to-end travel planning.
- Implemented AI-powered recommendation and optimization algorithms for personalized itineraries, budget allocation, and route planning, reducing manual planning effort by 70%.
- Designed a modular, scalable backend with REST APIs and dynamic integrations, delivering real-time travel insights, interactive maps, PDF itinerary generation, and AI chatbot assistance.

AI Personal Companion | *Python, Speech Recognition, Coqui TTS*

Feb 2025 – Apr 2025

- Developed a voice-enabled AI assistant using Python, Google Speech Recognition API, and Coqui TTS, enabling real-time speech recognition and natural voice synthesis.
- Implemented voice-command processing to trigger assistant actions and generate context-aware responses through the conversational pipeline.
- Integrated a VTuber anime character to provide real-time visual responses, creating an interactive conversational experience where the character responds to user voice commands.
- Optimized audio processing and inference workflows, improving response speed, speech clarity, and overall conversational quality.
- Implemented an end-to-end conversational AI framework that enhanced user interaction through intelligent voice command processing and natural language responses.

Heart Disease Predictor | *Python, TensorFlow, VGG16, CNN, Streamlit*

Sep 2024 – Dec 2024

- Developed a VGG16-based deep learning model for cardiovascular disease image classification using transfer learning.
- Built an image preprocessing pipeline involving normalization and augmentation to improve model robustness.
- Leveraged TensorFlow, CNNs, and feature extraction techniques to train and evaluate the classification model, achieving 87% classification accuracy.
- Built an interactive web interface using HTML, CSS, and JavaScript for uploading medical images and displaying model predictions.

Certifications

Intermediate Machine Learning

Kaggle — Certificate

Introduction to Deep Learning

Kaggle — Certificate

Machine Learning

NPTEL — IIT Madras

Oracle Cloud Infrastructure 2025 AI Foundations Associate

Oracle — Certificate

Oracle Certified Foundations Associate – Agentic AI certification

Oracle — Certificate

Technical Skills

Programming Languages: Python, C++

AI & Machine Learning: Scikit-learn, TensorFlow/Keras, Feature Engineering, Model Evaluation

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization

Frameworks & Tools: FastAPI, Streamlit, Git, GitHub, Jupyter Notebook, Google Colab

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, SQL